

Lecture 1: Basic Set Theory

Junnan Zhang

Paula and Gregory Chow Institute for Studies in Economics
Xiamen University

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Topics to be covered today

- ▶ Sets
- ▶ Functions
- ▶ Real Numbers

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- ▶ Set theory is the language of all mathematics
- ▶ In economics, sets are everywhere: consumption sets, production sets, sets of players, sets of equilibria, etc.

Definition 1

- ▶ A set is, informally speaking, a collection of objects, which we call **elements**.
- ▶ If x is an element of the set S , we write $x \in S$. We also say x **belongs to** S .
If x is not an element of S , we write $x \notin S$.

We use curly brackets when describing a set. We can list all its elements:

- ▶ $\{1, 2, 3\}$
- ▶ $\{1, 2, \{1, 3\}\}$

Or we can specify the properties that its elements satisfy:

- ▶ $\{x : x \text{ is irrational}\}$: the set of all x such that x is irrational
- ▶ $\{n \in \mathbb{N} : n^2 < 7\}$: the set of all natural numbers n such that $n^2 < 7$

Definition 2

- ▶ We say two sets A and B are **equal** if they have the same elements, and we write $A = B$.
- ▶ An **empty set** is a set with no elements and it is denoted by $\emptyset$.
- ▶ If $\forall x \in A, x \in B$, we say A is a **subset** of B and we write $A \subset B$.
- ▶ The set of all subsets of A is called the **power set** of A and is denoted by $\mathcal{P}(A)$.

We usually prove $A = B$ by proving $A \subset B$ and $B \subset A$.

- ▶ $\{\emptyset\}$ is not an empty set because $\emptyset \in \{\emptyset\}$
- ▶ $\emptyset \subset A$ for all sets A
- ▶ If $A = \{1, 2\}$, then $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$
- ▶ $\emptyset \in \mathcal{P}(A)$ for all sets A

Proposition 3

Set inclusion is transitive: if $A \subset B$ and $B \subset C$, then $A \subset C$.

Proof:

Definition 4

- ▶ The **union** of A and B is defined by

$$A \cup B := \{x : x \in A \text{ or } x \in B\}$$

- ▶ The **intersection** of A and B is defined by

$$A \cap B := \{x : x \in A \text{ and } x \in B\}$$

- ▶ Two sets A and B are **disjoint** if $A \cap B = \emptyset$

Union and Intersection of Sets

Venn Diagram

Definition 5

- ▶ If $\mathcal{F}$ is a family of sets, the union of all sets in $\mathcal{F}$ is defined by

$$\bigcup \mathcal{F} := \{x : x \in A \text{ for at least one } A \in \mathcal{F}\}$$

- ▶ Similarly, the intersection of all sets in $\mathcal{F}$ is defined by

$$\bigcap \mathcal{F} := \{x : x \in A \text{ for every } A \in \mathcal{F}\}$$

- ▶ If $\mathcal{F} = \{A_\lambda : \lambda \in J\}$, then

$$\bigcup \mathcal{F} = \bigcup_{\lambda \in J} A_\lambda \qquad \bigcap \mathcal{F} = \bigcap_{\lambda \in J} A_\lambda$$

Proposition 6

Let A, B, C and B_λ (for $\lambda \in J$) be sets. Then

$$A \cup B = B \cup A \quad A \cap B = B \cap A$$

$$A \cup (B \cup C) = (A \cup B) \cup C \quad A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \subset A \cup B \quad A \cap B \subset A$$

$$A \subset B \Leftrightarrow A \cup B = B \quad A \subset B \Leftrightarrow A \cap B = A$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap \bigcup_{\lambda \in J} B_\lambda = \bigcup_{\lambda \in J} (A \cap B_\lambda) \quad A \cup \bigcap_{\lambda \in J} B_\lambda = \bigcap_{\lambda \in J} (A \cup B_\lambda)$$

Prove some using the definitions

Definition 7

- ▶ The **difference** of A and B is defined by

$$A \setminus B \text{ (or } A - B) := \{x : x \in A \text{ and } x \notin B\}$$

- ▶ If U is a set that contains all objects being considered in a certain context (a universal set), then $U \setminus A$ is called the **complement** of A in U , and is denoted by A^c

Examples:

- ▶ If $U = \mathbb{R}$ and $A = (0, 1)$, then $A^c = (-\infty, 0] \cup [1, +\infty)$
- ▶ If $U = [0, 1]$ and $A = (0, 1)$, then $A^c = \{0, 1\}$

Proposition 8 (De Morgan's Laws)

Let A , B , and A_λ (for $\lambda \in J$) be sets. Then

$$(A \cup B)^c = A^c \cap B^c \qquad (A \cap B)^c = A^c \cup B^c$$

$$\left(\bigcup_{\lambda \in J} A_\lambda \right)^c = \bigcap_{\lambda \in J} A_\lambda^c \qquad \left(\bigcap_{\lambda \in J} A_\lambda \right)^c = \bigcup_{\lambda \in J} A_\lambda^c$$

Demonstrate using Venn diagrams

Definition 9

- ▶ An **ordered pair** whose first member is x and second member is y is denoted by (x, y)
- ▶ The basic property of ordered pairs is that $(x, y) = (a, b)$ iff $x = a$ and $y = b$
- ▶ The Cartesian product of two sets A and B is defined by

$$A \times B := \{(a, b) : a \in A, b \in B\}$$

- ▶ We can also define n -tuples $(a_1, a_2, \dots, a_n)$. Then the Cartesian product of n sets is defined by

$$\prod_{i=1}^n A_i := \{(a_1, a_2, \dots, a_n) : a_i \in A_i, \forall i = 1, 2, \dots, n\}$$

Cartesian Product

Examples

► $\{1, 2\} \times \{3, 4\} = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$

► $\mathbb{R}^n = \underbrace{\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}}_{n \text{ times}}$

- ▶ Natural numbers: $\mathbb{N} = \{1, 2, \dots\}$
- ▶ Integers: $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$
- ▶ Rational numbers: $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z}, q \neq 0\}$
- ▶ Real numbers: $\mathbb{R}$
- ▶ Complex numbers: $\mathbb{C}$

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What is a Function?

Definition 10

Let X and Y be two sets. A **function** or **map** or **mapping** f from X to Y associates to every element $x \in X$ an element $y \in Y$, denoted by $f(x)$. We write

$$f: X \rightarrow Y$$

$$x \mapsto f(x)$$

We call X the **domain** of f and Y the **codomain** of f .

We call x an **argument** of $f(x)$, and $f(x)$ the **image** of f at x .

Definition 11

- ▶ Let $S \subset X$. The **image** of S under f is defined by $f(S) = \{f(x) : x \in S\}$.
- ▶ The image of X is called the **range** of f .
- ▶ Let $T \subset Y$. The **inverse image** of T under f is defined by $f^{-1}(T) := \{x : f(x) \in T\}$.
- ▶ We say f is **one-one** or **injective** if for every $y \in Y$, there is at most one $x \in X$ such that $y = f(x)$.
- ▶ We say f is **onto** or **surjective** if for every $y \in Y$, there exists $x \in X$ such that $y = f(x)$.
- ▶ If f is both one-one and onto, we say f is **bijective**.

Image and Inverse Image

Examples

Proposition 12

$$f(C \cup D) = f(C) \cup f(D) \qquad f\left(\bigcup_{\lambda \in J} A_\lambda\right) = \bigcup_{\lambda \in J} f(A_\lambda)$$

$$f(C \cap D) \subset f(C) \cap f(D) \qquad f\left(\bigcap_{\lambda \in J} A_\lambda\right) \subset \bigcap_{\lambda \in J} f(A_\lambda)$$

$$f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D) \qquad f^{-1}\left(\bigcup_{\lambda \in J} A_\lambda\right) = \bigcup_{\lambda \in J} f^{-1}(A_\lambda)$$

$$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D) \qquad f^{-1}\left(\bigcap_{\lambda \in J} A_\lambda\right) = \bigcap_{\lambda \in J} f^{-1}(A_\lambda)$$

$$(f^{-1}(A))^c = f^{-1}(A^c) \qquad \text{no general results for } f$$

$$A \subset f^{-1}(f(A))$$

Definition 13

- ▶ If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, then the **composition** function $g \circ f : X \rightarrow Z$ is defined by

$$(g \circ f)(x) = g(f(x))$$

- ▶ The **identity** function on X is defined by $Id_X(x) = x$ for all $x \in X$.
- ▶ We say $f : X \rightarrow Y$ is **invertible** if there exists $g : Y \rightarrow X$ such that $f \circ g = Id_Y$ and $g \circ f = Id_X$. In this case, g is the **inverse** of f and is denoted by f^{-1} .

- ▶ Example: $f(x) = x^2$, $g(x) = \sin(x)$. Then $(g \circ f)(x) = \sin(x^2)$ and $(f \circ g)(x) = \sin^2(x)$.
- ▶ The composition operator is associative: $h \circ (g \circ f) = (h \circ g) \circ f$.

Proposition 14

f is bijective iff f is invertible.

Proof: Suppose that f is bijective. Then, for every $y \in Y$, there is a unique $x \in X$ such that $f(x) = y$. This allows us to define $g : Y \rightarrow X$ by $f(g(y)) = y$. Therefore, $f \circ g = Id_Y$. Now pick any $x \in X$ and let $y = f(x)$. The definition of g implies that $g(y) = x$. Therefore, $g(f(x)) = x$ and thus $g \circ f = Id_X$.

Suppose f is invertible. Then for any $y \in Y$ there is an $x \in X$ given by $x = f^{-1}(y)$ such that $f(x) = f(f^{-1}(y)) = Id_Y(y) = y$. Now suppose there is another $x' \in X$ such that $f(x') = y$. Then $f^{-1}(f(x')) = Id_X(x') = x'$. Since $f^{-1}(f(x')) = f^{-1}(y) = x$, $x = x'$. Therefore, f is bijective.

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- ▶ Most functions we encounter in economics are real-valued functions
- ▶ It is important to know some basic properties of the real numbers
- ▶ It is also key to the study of convergence later on
- ▶ We take the familiar algebraic and order properties of the real numbers as given; their formal statement is included in the appendix.

Definition 15

If S is a set of real numbers, then

- 1 a is an **upper bound** for S if $x \leq a$ for all $x \in S$;
- 2 b is the **least upper bound** (or **l.u.b.** or **supremum** or **sup**) for S if b is an upper bound and $b \leq a$ whenever a is an upper bound.
- 3 We call b a **maximum** of S if b is the l.u.b. for S and $b \in S$.

We can similarly define **lower bound**, **greatest lower bound** (or **g.l.b.** or **infimum** or **inf**), and **minimum**.

A set is **bounded above** if it has an upper bound and is **bounded below** if it has a lower bound.

A12 Least Upper Bound Completeness Axiom

Suppose S is a nonempty set of real numbers that is bounded above. Then S has a l.u.b in $\mathbb{R}$.

Corollary 16

Suppose S is a nonempty set of real numbers that is bounded below. Then S has a g.l.b in $\mathbb{R}$.

Proposition 17

Suppose S is a nonempty set of real numbers. Then b is the supremum of S iff

- ① $x \leq b$ for all $x \in S$, and
- ② for each $\epsilon > 0$, there exists $x \in S$ such that $x > b - \epsilon$.

Proof:

Theorem 18

The set $\mathbb{N}$ is not bounded above.

Proof: Suppose $\mathbb{N}$ is bounded above. By the Completeness Axiom, there is a supremum $b \in \mathbb{R}$ such that $n \leq b$ for all $n \in \mathbb{N}$. Since $n + 1 \in \mathbb{N}$, $n + 1 \leq b$ and thus $n \leq b - 1$ for all $n \in \mathbb{N}$. This is a contradiction since b is no longer the least upper bound of $\mathbb{N}$.

Theorem 19

- 1 For any $x, y \in \mathbb{R}$, if $x < y$ then there exists a rational (irrational) number r such that $x < r < y$.
- 2 For any $x \in \mathbb{R}$ and $\epsilon > 0$, there exists a rational (irrational) number r such that $0 < |r - x| < \epsilon$.

The Archimedean Property

Exercise

Prove that the supremum of $\{x \in \mathbb{R} : x < 3\}$ is 3.

Proof:

Definition 20

Let $f : X \rightarrow Y$ be a **real-valued** function, that is, $Y \subset \mathbb{R}$.

Define upper bounds, the supremum, the maximum of f as the upper bounds, supremum, and maximum of its range $f(X) \subset \mathbb{R}$. The arguments x such that $f(x)$ is the maximum are called the **maximizers** of f .

We say f is **bounded** if its range is bounded.

Definition 21

If $X \subset \mathbb{R}$, then we say

- ▶ f is **increasing** if $x \leq x' \Rightarrow f(x) \leq f(x')$
- ▶ f is **decreasing** if $x \leq x' \Rightarrow f(x) \geq f(x')$
- ▶ f is **monotonic** if it is increasing or decreasing
- ▶ f is **strictly increasing** if $x < x' \Rightarrow f(x) < f(x')$
- ▶ f is **strictly decreasing** if $x < x' \Rightarrow f(x) > f(x')$

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Definition 22

- ▶ Two sets X and Y are **equivalent** if there exists a bijection $f : X \rightarrow Y$. We write $X \sim Y$.
- ▶ With every set A we associate a **cardinal number** or **cardinality** to indicate the “number” of elements in A . Two sets have the same cardinality if they are equivalent.
- ▶ Y has higher cardinality than X if there exists an injection from X into Y .

Definition 23

- ▶ A set is finite if it is empty or it is equivalent to the set $\{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$. In this case, it has cardinality n . A set is **infinite** if it is not finite.
- ▶ A set is **denumerable** (or countably infinite) if it is equivalent to $\mathbb{N}$. A set is **countable** if it is denumerable or finite. If a set is denumerable, we say it has cardinality d .
- ▶ A set is **uncountable** if it is not countable. If a set is equivalent to $\mathbb{R}$, we say it has cardinality c .

Proposition 24

- ① *Any subset of a countable set is countable.*
- ② *The set of rational numbers $\mathbb{Q}$ is denumerable.*
- ③ *The set of even numbers is equivalent to $\mathbb{N}$. The set $(0, 1)$ is equivalent to $\mathbb{R}$.*
- ④ *The set $\mathbb{N}$ and $(0, 1)$ are not equivalent.*
- ⑤ *The product of two countable sets is countable.*
- ⑥ *The cardinality of $\mathbb{R}^n$ is c .*
- ⑦ *The product of two sets of cardinality c has cardinality c .*
- ⑧ *The union of a countable family of countable sets is countable.*

The **Real Number System** is a set of objects called **Real Numbers** together with two binary operations called **addition** (+) and **multiplication** ($\times$), a binary relation called **less than** ($<$), and two elements called **zero** (0) and **unity** (1), that satisfies a set of axioms:

► properties of addition

A1 $a + b = b + a$

A2 $(a + b) + c = a + (b + c)$

A3 $a + 0 = 0 + a = a$

A4 there is exactly one real number, denoted by $-a$ such that $a + (-a) = (-a) + a = 0$

► properties of multiplication

A5 $a \times b = b \times a$

A6 $(a \times b) \times c = a \times (b \times c)$

A7 $a \times 1 = 1 \times a = a$, and $1 \neq 0$

A8 if $a \neq 0$ then there is exactly one real number, denoted by a^{-1} such that $a \times a^{-1} = a^{-1} \times a = 1$

► the distributive property

A9 $a(b + c) = ab + ac$

► Axioms A1 - A9 are called algebraic axioms

Definition 25

- ▶ Define **subtraction** by $a - b = a + (-b)$
- ▶ Define **division** by $a/b = a \times b^{-1}$ for $b \neq 0$

Theorem 26

- 1 If $a + c = b + c$, then $a = b$.
- 2 If $ac = bc$ and $c \neq 0$, then $a = b$.

Proof: Since $a + c = b + c$, $(a + c) + (-c) = (b + c) + (-c)$. It follows from A2 that $a + (c + (-c)) = b + (c + (-c))$. By A4, $a + 0 = b + 0$. By A3, $a = b$. The second claim can be proved in a similar way.

Theorem 27

If $c \neq 0$ and $d \neq 0$, then

- ① $a0 = 0, -(-a) = a$
- ② $(c^{-1})^{-1} = c, (-1)a = -a$
- ③ $a(-b) = -(ab) = (-a)b$
- ④ $(-a) + (-b) = -(a + b)$
- ⑤ $(-a)(-b) = ab$
- ⑥ $(a/c)(b/d) = (ab)/(cd)$
- ⑦ $(a/c) + (b/d) = (ad + bc)/(cd)$

Prove some using Axioms A1-A9

- ▶ The Order Axioms: there is a subset P of the real numbers, called the set of **positive numbers** such that

A10 For any real number a , exactly one of the following holds:

$$a = 0 \quad \text{or} \quad a \in P \quad \text{or} \quad -a \in P$$

A11 If $a \in P$ and $b \in P$, then $a + b \in P$ and $ab \in P$

- ▶ The “Less Than” Relation: define $a < b$ to mean $b - a \in P$, $a \leq b$ to mean $b - a \in P$ or $b - a = 0$, $a > b$ to mean $a - b \in P$, and $a \geq b$ to mean $a - b \in P$ or $a = b$.
- ▶ $a < b$ iff $b > a$ and $a \leq b$ iff $b \geq a$

Theorem 28

- ① $a < b$ and $b < c$ implies $a < c$
- ② exactly one of $a < b$, $a = b$, and $a > b$ is true
- ③ $a < b$ implies $a + c < b + c$
- ④ $a < b$ and $c > 0$ implies $ac < bc$
- ⑤ $a < b$ and $c < 0$ implies $ac > bc$
- ⑥ $0 < 1$ and $-1 < 0$
- ⑦ $a > 0$ implies $1/a > 0$
- ⑧ $0 < a < b$ implies $0 < 1/b < 1/a$

Similar results hold for $\leq$

Prove some using Axioms A1-A11